

S. Maitrey

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Employment History

2024 – Present 📖 **Project Fellow**, Exoplanets and Planetary Formation Group, School of Earth and Planetary Sciences, National Institute of Science Education and Research, Bhubaneswar.

Education

2019 – 2024 📖 **Integrated Master of Science (Physics)** at School of Physical Sciences, National Institute of Science Education and Research, Bhubaneswar.
Thesis title: *Tracing the evolution of gas and dust at planetary formation scales using the Atacama Large Millimeter/submillimeter Array (ALMA)*
Supervisor: Dr. Liton Majumdar

Publications

Lead Author

1 S. Maitrey, L. Majumdar, V. Manilal, *et al.*, “Role of Diffusive and Non-Diffusive Grain-Surface Processes in Cold Cores: Insights from the PEGASIS Three-Phase Astrochemical Model,” *Accepted for publication in Astronomy & Astrophysics (A&A)*, Apr. 2025, *Accepted at A&A*. [DOI: 10.48550/arXiv.2504.18138](#). arXiv: 2504.18138 [astro-ph.GA].

Co-author

1 P. Rayalacheruvu, L. Majumdar, W. R. M. Rocha, *et al.*, “Expanding the Ice Inventory of NGC 1333 IRAS 2A with INDRA using JWST Observations: Tracing Organic Refractories and Beyond,” *Submitted to Astrophysical Journal Supplement Series (ApJS)*, Jun. 2025, *Under Review at ApJS*. arXiv: 2506.15358 [astro-ph.GA].

Projects

2023 – Present 📖 **Constraining the disk gas mass through well-established chemical tracers to constrain giant planet formation**
Development of an exhaustive radiative transfer and self-consistent gas-grain thermochemical code to study planet-forming regions and commenting on material available for the genesis of giant planets in later stages of planet formation.
Supervisor: Dr. Liton Majumdar (School of Earth and Planetary Sciences, NISER).

2025 📖 **Exploring effects of cosmic-ray ionization in protoplanetary disks**
Implemented various prescriptions to model cosmic-ray ionization attenuation for the disk HD163296 to fit ALMA observations and explored how different modes of ionization (X-ray, short-lived radionuclides, Lyman-Alpha and cosmic-rays) can influence the disk chemistry. This work is currently under preparation.
Supervisor: Dr. Liton Majumdar (School of Earth and Planetary Sciences, NISER).

Projects (continued)

- 2024 ■ **Development of a novel pipeline to explore and constrain ice inventory from JWST observations**
Planned and designed the initial stages of the algorithm to constrain ice column densities from JWST observational data. This work is currently under review at *The Astrophysical Journal Supplement Series (ApJS)*.
Supervisor: Dr. Liton Majumdar (School of Earth and Planetary Sciences, NISER).
- 2022 – 2024 ■ **Development of a three-phase gas-grain astrochemical code with state-of-the-art chemistry.**
Lead developer of a comprehensive three-phase gas-grain astrochemical model with isotope and spin-state chemistry designed to study chemical evolution in planet-forming regions, from molecular clouds and protostars to protoplanetary disks. This work has been accepted at *Astronomy & Astrophysics (A&A)*.
Supervisor: Dr. Liton Majumdar (School of Earth and Planetary Sciences, NISER).
- 2024 ■ **Development of an easy-to-use Python wrapper to ODEPACK solvers.**
Developed a Python wrapper to the `DLSODE` family of ordinary differential equation solvers for stiff systems and problems involving sparse Jacobians from the Fortran library ODEPACK.
- 2023 ■ **Modeling the interior evolution of rocky exoplanets using Machine Learning.**
Built a model which can predict the evolution of terrestrial exoplanet interiors using recurrent neural networks (RNN) from the `scikit-learn` library, by using outputs from the `VPLANET` software library as training data.
Supervisors: Dr. Subhankar Mishra (School of Computer Sciences, NISER) and Dr. Liton Majumdar (School of Earth and Planetary Sciences, NISER).

Skills

- Languages ■ Strong reading, writing and speaking competencies for English (**TOEFL Score 111 / 120**) and Hindi.
- Programming ■ Python, Fortran, Rust, C, Julia and $\text{\LaTeX}$.
- Codes and models used ■ `rac-2d` (a thermochemical model for protoplanetary disks), `RADMC-3D` (radiative transfer code), `NAUTILUS` (gas-grain astrochemical code), `Optool` (Tool to create opacities for astrophysical dust), `RADEX` (program to perform non-LTE line transfer through escape probability formulation), `RADlite` (a fast 2D/3D raytracer for infrared lines), `LIME` (Line Modeling Engine, a 3-D radiative transfer code for modelling far-IR and sub-mm observations) `Meudon PDR` (code to study physics and chemistry in diffuse clouds), `healpy`. Contributed to `astropy/astroquery`.

Miscellaneous Experience

Conferences and Workshops

- 2022 ■ **Selected** for the summer school through the Summer Programme organised by the Indian Institute of Astrophysics (IIA), Bangalore
- 2023 ■ Attended the **Strange New Worlds: The Exploration of Exoplanets** conference on exoplanetary science at IISER Pune during August 17-19, 2023 at Pune.
- 2019 ■ Attended RAD@Home Radio Astronomy Workshop at IISER Berhampur, India.

Miscellaneous Experience (continued)

Talks and Posters

- 2023  **Presented and Won the 4th best poster award** for a contributed poster at the **Strange New Worlds: The Exploration of Exoplanets** conference on exoplanetary science at IISER Pune during August 17-19, 2023.

Awards and Achievements

- 2019  **DISHA (DAE Incentive Scheme for Holistic Science Education and Augmentation)** – 5-year scholarship for undergraduate studies, by Department of Atomic Energy (DAE), Government of India.
- 2018  **Qualified** for *Technothon Mains*, organized by IIT Guwahati.
- 2017  **Recipient of the National Talent Search Examination (NTSE) Scholarship** organized by the National Council of Educational Research and Training (NCERT), New Delhi.
- 2016  **Qualified** for the Indian National Mathematical Olympiad (INMO), conducted for the selection for the International Mathematical Olympiad by the Homi Bhabha Centre for Science Education (HBCSE) and Tata Institute of Fundamental Research (TIFR).

Extra-curricular

- 2019 – 2024  **As part of the NISER Astronomy Club:**
- Contributed to various astrophotography sessions, especially smartphone astrophotography.
 - Conducted online astronomy quizzes through remote interactions.
 - Served as one of the principal content editors for the NISER Astronomy Club's biannual magazine *Kshitij*.
 - Served as one of the content writers for the monthly newsletter published by the NISER Astronomy Club.
- 2021  Participated and **won 2nd** prize at the Intra-NISER Table Tennis tournament.
- 2019  Tutored senior secondary school students of JNV Dhenkanal as a part of AVANTI mentorship program.
-  **Won the first prize** in the team category, *NISER Quizzone Club*.
- 2017  Notable representation of my school at the CBSE Science Exhibition, for building a “*A Working model of energy efficient highways*”, CBSE.
- 2015-2018  Participated in various high school level quizzes organized by CBSE like the CBSE Cultural Heritage Quizzes over my school tenure.